Solution Requirements

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The administrator can securely access and manage the application. Users can use the prediction system without logging in.	Medium
2	Authorization Levels	The administrator has permission to manage the machine learning model and application. Users can only enter applicant details and view prediction results.	Medium
3	External Interfaces	The system uses CSV datasets for training, a trained Random Forest model (model.pkl) for prediction, and a Flask web interface for user interaction.	High
4	Transaction Processing	The application accepts applicant details, preprocesses the data, predicts credit card approval, and displays the result with a confidence score.	High

5	Reporting	The system displays the prediction result (Approved/Rejected) with the confidence score. Model comparison results are stored in model_comparison.csv.	High
6	Business Rules	The prediction is based on applicant details such as income, age, employment status, education, occupation, family members, housing type, and ownership details. All mandatory fields must be entered before prediction.	High
7	Compliance to Laws or Regulations	The application is developed for educational purposes and user information should be processed securely and used only for prediction.	Medium
8	Other	The system provides a simple, responsive, and user-friendly interface for predicting credit card approval.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should generate the prediction quickly after the user submits the application.	Prediction displayed within 2 seconds
2	Scalability	The application should support multiple prediction requests without affecting performance.	Supports more than 100 prediction requests per hour
3	Security & Data Privacy	User information should be processed securely with proper input validation and should not be stored permanently unless required.	Secure input validation and protected data processing
4	Reliability & Availability	The application should remain available whenever users access it for prediction.	99% system availability during usage
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use for all users.	User-friendly interface with clear labels and navigation
6	Maintainability	The project should be modular so that new machine learning models or additional features can be added easily in the future.	Easy to maintain and update